Name_____ Date_____ Period_____

LESSON 7-4 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Comparing Numbers on a Number Line

Maryland Standard 6.NS.C.7a · Learning target – I can compare two rational numbers using their positions on a number line.

REMEMBER

- Numbers get **greater to the right** and **less to the left**.
- Any positive number is greater than any negative number.
- −2 > −8 because −2 is farther right. **<** means is less than; **>** means is greater than.

WORKED EXAMPLE

Compare −3.5 and −1.5. Use < or >.

Step 1 −1.5 is to the right of −3.5 on the number line.

Step 2 Right means greater.

Answer **−1.5 > −3.5**

WORD BANK

greater than (>) – farther right on the number line

less than (<) – farther left on the number line

compare – decide which number is greater

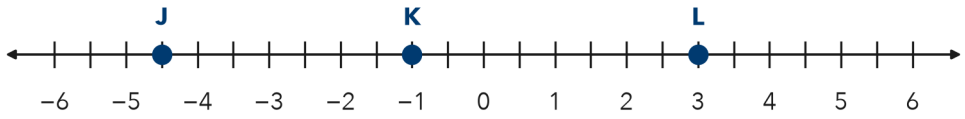
PRACTICE

show your work · label units

1. Use the number line. Write < or > in each blank.

Step 1 Find the number for J, K, and L.

Step 2 The point farther right is greater.



a. J ___ K **b.** L ___ K **c.** J ___ L

2. Write < or > in each blank.

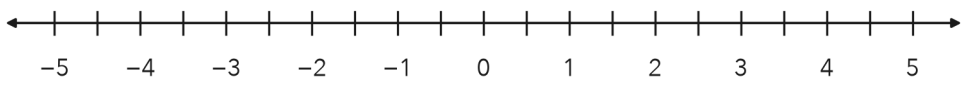
Step 1 Picture both numbers on a number line.

Step 2 Write > if the first number is farther right. Write < if it is farther left.

a. −8 ___ −3 **b.** 0 ___ −6 **c.** −2.5 ___ −2.05

d. 4 ___ −10 **e.** −1/2 ___ −3/4 **f.** −7 ___ −7.1

3. Plot −3.5 and −0.5 on the number line. Then write a true comparison using <. _____



4. Golf. In golf, the **lower** score wins. Tanya scored −3. Malik scored −5.

a. Compare the scores using < or >. _____

b. Who won? _____

5. Cold Days. Monday's low was −7 degrees. Tuesday's low was −2 degrees. Which day was colder? Write a comparison using < or >.

Answer _____

6. Reasoning. Is −6 < |−6| true? Explain how you know.

Frame *|−6| = ___. −6 is to the ___ of 6 on the number line, so −6 < |−6| is ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask someone at home: which is colder, −4 degrees or −9 degrees? Why?

Name_____ Date_____ Period_____

LESSON 7-4 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Ordering Integers and Rational Numbers

Maryland Standard 6.NS.C.7b · Learning target – I can order integers and rational numbers from least to greatest.

REMEMBER

- Place the numbers on a number line. **Least** is farthest left; **greatest** is farthest right.
- Change fractions to decimals to compare: $-3/4 = -0.75$.
- For negative numbers, the one farther from 0 is **less**.

WORKED EXAMPLE

Order -1.2, 0.5, -2, and $-1/2$ from least to greatest.

Step 1 Write $-1/2$ as -0.5.

Step 2 Read the number line from left to right.

Answer **$-2, -1.2, -1/2, 0.5$**

WORD BANK

order – put numbers in a row
least – farthest left
greatest – farthest right
deepest – the least (most negative) number

PRACTICE

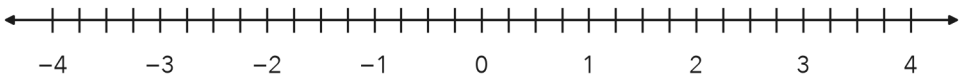
show your work · label units

1. Plot each number. Then write them in order from least to greatest: 1.5, −2.5, −3/4, 2 1/4, −3.25

Step 1 Change −3/4 and 2 1/4 to decimals.

Step 2 Plot each number.

Step 3 Read the points from left to right.



Order_____

2. Order each set from least to greatest.

Step 1 Put the negative numbers first. The one farthest from 0 goes first.

Step 2 Then write 0 and the positive numbers.

a. −6, 4, −9, 0 _____

b. −0.3, −3, −0.03 _____

3. Circle the greatest number in each set.

a. −5, −2, −8

b. −1.4, −1.04, −1.44

c. −2/3, −1/2, −3/4

d. 0, −0.1, −10

4. **Ocean Animals.** A shark swims at −45.5 m. A turtle is at −12 m. A dolphin is at −30.25 m. A jellyfish is at −8.75 m. List them from **deepest** to **closest to the surface**.

Answer_____

5. **Find the Error.** Leo ordered −1/2, −0.4, and −0.6 from least to greatest as −0.4, −1/2, −0.6. Explain his mistake and write the correct order.

Frame *Leo is wrong because for negative numbers, farther from 0 is ___. The correct order is ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. Which list is ordered from least to greatest?

A −5, −8, 0, 3 **B** −8, −5, 0, 3 **C** 0, −5, −8, 3 **D** 3, 0, −5, −8

7. Which number is greatest?

A −2.7 **B** −2.07 **C** −2.77 **D** −2.2

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

AT HOME List the low temperatures for three cold days with someone at home. Order them from coldest to warmest.